

20260618

Business Intelligence

- 1] A BI analyst wants to cluster small business customers according to

X_1 = monthly software spend (hundreds EUR)

X_2 = number of support tickets

- The 4 customers are:

$$A = (1, 1), \quad B = (2, 1), \quad C = (6, 5), \quad D = (7, 5)$$

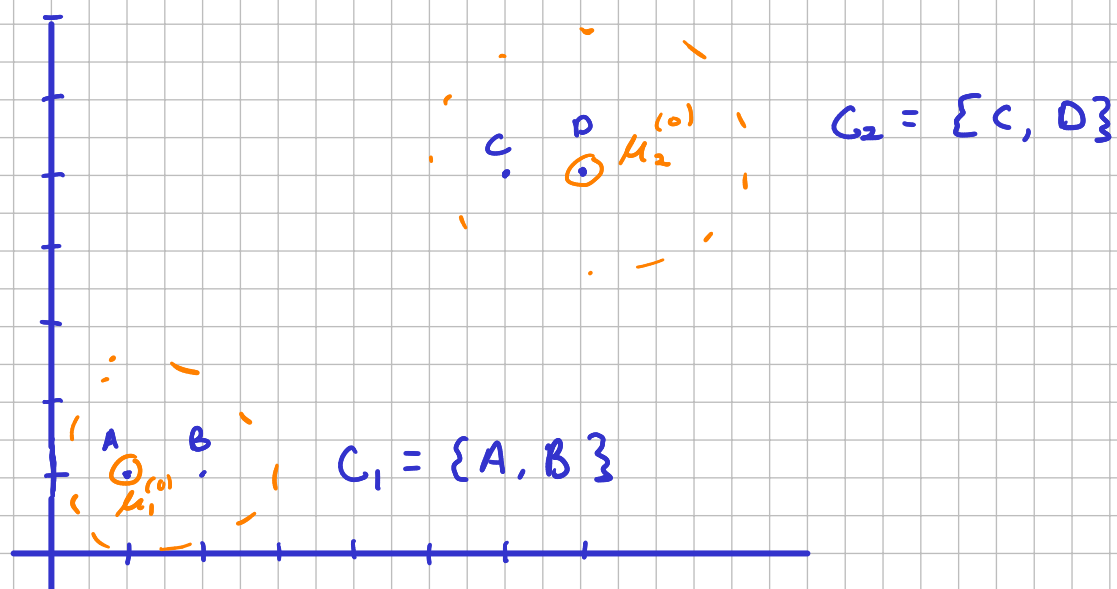
- The initial K-means centroids

$$\mu_1^{(0)} = (1, 1) \quad \mu_2^{(0)} = (7, 5)$$

- a) Compute the Euclidean distance from each customer to the nearest centroid.

Customer	$\mu_1^{(0)} = (1, 1)$	$\mu_2^{(0)} = (7, 5)$
$A = (1, 1)$	$d = \sqrt{(1-1)^2 + (1-1)^2}$ $= \sqrt{0} = 0$	$d = \sqrt{(1-7)^2 + (1-5)^2}$ $= \sqrt{36 + 16} \approx 7.211$
$B = (2, 1)$	$d = \sqrt{(2-1)^2 + (1-1)^2}$ $= \sqrt{1} = 1$	$d = \sqrt{(2-7)^2 + (1-5)^2}$ $= \sqrt{25 + 16} \approx 6.403$
$C = (6, 5)$	$d = \sqrt{(6-1)^2 + (5-1)^2}$ $= \sqrt{25 + 16} \approx 6.403$	$d = \sqrt{(6-7)^2 + (5-5)^2}$ $= \sqrt{1} = 1$
$D = (7, 5)$	$d = \sqrt{(7-1)^2 + (5-1)^2}$ $= \sqrt{36 + 16} \approx 7.211$	$d = 0$

b) Assign each customer to the nearest centroid.



c) Compute the updated centroids

$$\mu_1^{(1)}: \quad A = (1, 1), \quad B = (2, 1)$$

$$\text{mean } x_1 \Rightarrow \frac{1 + 2}{2} = \frac{3}{2} = 1.5$$

$$\text{mean } x_2 \Rightarrow \frac{1 + 1}{2} = 1$$

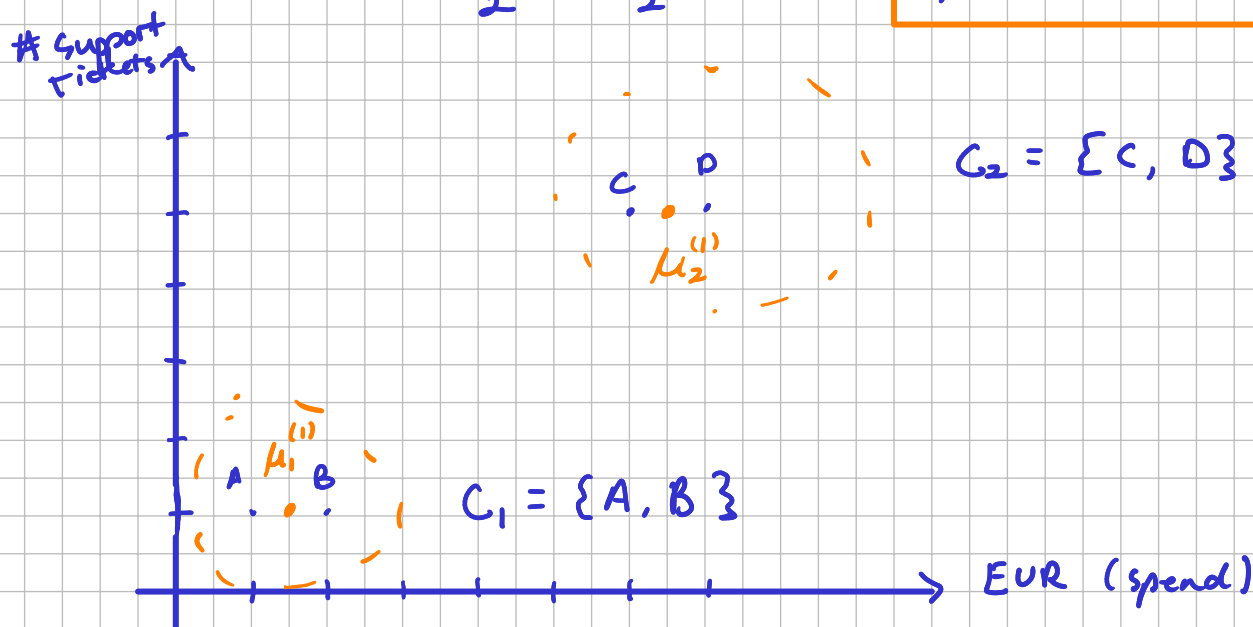
$$\mu_1^{(1)} = (1.5, 1)$$

$$\mu_2^{(1)}: \quad C = (6, 5), \quad D = (7, 5)$$

$$\text{mean } x_1 \Rightarrow \frac{6 + 7}{2} = \frac{13}{2} = 6.5$$

$$\text{mean } x_2 \Rightarrow \frac{5 + 5}{2} = \frac{10}{2} = 5$$

$$\mu_2^{(1)} = (6.5, 5)$$



d) Interpret the 2 clusters in CRM language.

- Cluster 1 (Customers A and B):

These customers are small-scale accounts. They are low-revenue generators, but require fewer support tickets and time.

- Cluster 2 (Customers C and D):

These customers are high-value accounts. They generate more revenue but require more support.

[2] An ERP system stores process observations with 2 variables:

X_1 = Processing time in minutes

X_2 = Number of manual corrections

The observations are:

$P_1 = (2, 1)$, $P_2 = (3, 1)$, $P_3 = (4, 2)$, $P_4 = (9, 6)$, $P_5 = (10, 7)$,
 $P_6 = (11, 7)$.

⇒ After one K-means iteration, the cluster assignment is:

$C_1 = \{P_1, P_2, P_3\}$, $C_2 = \{P_4, P_5, P_6\}$.

a) Compute the centroid of C_1 .

We know $|C_1| = 3$

Mean x_1 (Processing time):

$$\frac{2 + 3 + 4}{3} = \frac{9}{3} = 3$$

Mean x_2 (corrections):

$$\frac{1 + 1 + 2}{3} = \frac{4}{3} \approx 1.333$$

$$\mu_1 = (3, 1.333)$$

b) Compute centroid for C_2 .

Mean x_1 (Processing time):

$$\frac{9 + 10 + 11}{3} = \frac{30}{3} = 10$$

Mean x_2 (corrections):

$$\frac{6 + 7 + 7}{3} = \frac{20}{3} \approx 6.667$$

$$\mu_2 = (10, 6.667)$$

c) Compute the within-cluster sum of squares J .

$\Rightarrow$ To calculate the within-cluster sum of squares, we calculate the squared distance from each point to its assigned centroid and then add them together.

* Because it's a squared distance, we can eliminate the roots!

$$d = \left(\sqrt{(-)^2 + (-)^2} \right)^2 = (-)^2 + (-)^2$$

C_1 $\Rightarrow P_1 = (2, 1): \mu_1 = (3, 1.333)$

$$(2 - 3)^2 + (1 - 1.333)^2 = 1 + 0.111 = \underline{1.111}$$

$$P_2 = (3, 1)$$

$$(3 - 3)^2 + (1 - 1.333)^2 = 0 + 0.111 = \underline{0.111}$$

$$P_3 = (4, 2)$$

$$(4 - 3)^2 + (2 - 1.333)^2 = 1 + 0.445 = \underline{1.445}$$

C_2 $\Rightarrow \mu_2 = (10, 6.667)$

$$P_4 = (9, 6)$$

$$(9 - 10)^2 + (6 - 6.667)^2 = 1 + 0.445 = \underline{1.445}$$

$$P_5 = (10, 7)$$

$$(10 - 10)^2 + (7 - 6.667)^2 = 0 + 0.111 = \underline{0.111}$$

$$P_6 = (11, 7)$$

$$(11 - 10)^2 + (7 - 6.667)^2 = 1 + 0.111 = \underline{1.111}$$

$$\sum C_1 = 1.111 + 0.111 + 1.445 = 2.667$$

$$\sum C_2 = " " = 2.667$$

$$J = \sum_k = C_1 + C_2 = 2 \times 2.667 = 5.334$$

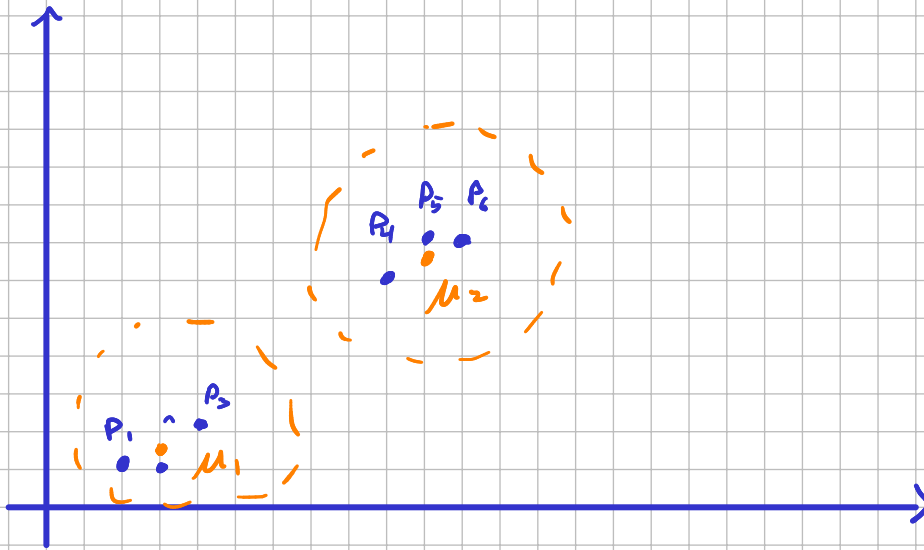
d) Explain what a high value of J would mean in a process-mining dashboard.

$\Rightarrow$ In a process-mining dashboard, the k-means objective J measures the total variance in the groups.

A high value of J means high heterogeneity.

$\Rightarrow$ The processes grouped into the same cluster are very different from one another!

$\Rightarrow$ Poor group separation! Increase # of clusters



8] A KNN classifier predicts whether a customer will buy an add-on service.

Training data:

X_1 = number of previous purchases

X_2 = website visits last week

Customer	x_1	x_2	Bought add-on
A	1	1	N
B	2	1	N
C	4	3	Y
D	5	4	Y
E	6	5	Y

A new customer has $x_{\text{new}} = (3, 2)$.

a) Compute the Euclidean distance between x_{new} and each training customer.

$$A = (1, 1): d = \sqrt{(1-3)^2 + (1-2)^2} \approx 2.236$$

$$B = (2, 1): d = \sqrt{(2-3)^2 + (1-2)^2} \approx 1.414$$

$$C = (4, 3): d = \sqrt{(4-3)^2 + (3-2)^2} \approx 1.414$$

$$D = (5, 4): d = \sqrt{(5-3)^2 + (4-2)^2} \approx 2.828$$

$$E = (6, 5): d = \sqrt{(6-3)^2 + (5-2)^2} \approx 4.243$$

b) Calculate the $K=3$ nearest neighbors.

$\Rightarrow$ To identify the 3 NNs, we sort the distances found in part (a) in ascending order:

$$\left. \begin{array}{l} 1. d_B = 1.414 \\ 2. d_C = 1.414 \\ 3. d_A = 2.236 \\ \vdots \end{array} \right\} \text{We can stop here.}$$

c) Predict the class using majority vote.

1. B : N

2. C : Y

3. A : N

Majority vote : N

$$p(X_{\text{new}} \in N) = \frac{2}{3} \approx 66\%$$

$$p(X_{\text{new}} \in Y) = \frac{1}{3}$$

Prediction ($\hat{y}$) : The new customer will NOT buy the add-on.

d) Explain the prediction in sales-dashboard language.

=> The dashboard should flag this customer as a "cold" lead. This means we should not spend resources on a follow-up conversation.

- 15] A linear SVM is trained to separate 2 customer classes:
- +1 Premium customers
 - 1 standard customers

The model is

$$f(x) = w^T x + b, \quad w = (2, 1), \quad b = -5.$$

For the customers

$$A = (1, 1), \quad B = (2, 2), \quad C = (3, 1), \quad D = (4, 2),$$

a) Compute $f(x)$ for each customer.

$$f(x) = (2)x_1 + (1)x_2 - 5$$

$$\Rightarrow f(A) = 2(1) + 1(1) - 5 = -2$$

$$f(B) = 2(2) + 1(2) - 5 = 1$$

$$f(C) = 2(3) + 1(1) - 5 = 2$$

$$f(D) = 2(4) + 1(2) - 5 = 5$$

b) Assign the predicted class to each customer using $\text{sgn}(f(x))$.

$$\Rightarrow A: \text{sgn}(f(A)) = \text{sgn}(-2) = -1 \quad (\text{standard})$$

$$B: \text{sgn}(f(B)) = \text{sgn}(1) = +1 \quad (\text{Premium})$$

$$C: \text{sgn}(f(C)) = \text{sgn}(2) = +1 \quad (\text{Premium})$$

$$D: \text{sgn}(f(D)) = \text{sgn}(5) = +1 \quad (\text{Premium})$$

c) Write the equation for the separating line.

$\Rightarrow$ The separating line (decision boundary) is where the decision function equals exactly zero.

$$f(x) = w^T x + b = 0$$

$$= (2)x + (1)y - 5 = 0$$

* plugged in $w = (2, 1)$, $b = -5$

$\Rightarrow$ We just solve for y :

$$y = -2x + 5$$

d) Compute $\|w\|$

$$\|w\| = \sqrt{w_1^2 + w_2^2} = \sqrt{2^2 + 1^2} = \sqrt{5} \approx 2.236$$

e) Explain what the sign of $f(x)$ means in a BI classification language.

$\Rightarrow$ In BI language, $\text{sgn}(f(x))$ acts as a segmentation classifier.

+1 Classifies the customer into a high-value segment (Premium)

-1 Classifies the customer into a lower-value segment (standard).